ICDETECTOR EXPORTS · FULL TECHNICAL REPORT

Technical report — Patrones sintéticos

What ICdetector recorded, what CellCase checks in the data and what is worth reviewing. It does not attribute attacks.

Case	Patrones sintéticos — Reproduce la forma de los patrones de un expediente real, con datos inventados
Capture device	Teléfono sintético (sin dispositivo real) — declared by the user; not verified
Data origin	Synthetic test data
Capture time zone	Europe/Madrid — declared and confirmed assumption
Scope	5 of 5 files imported: Observation history, Radio: service state events, Topology, Stable-Site, Geometry
Period	2026-09-24 – 2026-10-02 (history, phone's local time)
Generated	2026-10-02 08:05:00 UTC · CellCase 0.2.0 · episodes/7
Readers and rules	readers/2.4.1 · cellcase-rules/3 · episodes/7 · cross/4
Report language	English (en-GB); the language only changes the presentation, not the analysis

Notice
Synthetic test data: they do not describe any real phone, network or place.

1. Summary

> **Assessment for the spoofing question (the same as in the plain-language report)**
> Observations that need more data. Technical interest (review priority) is a separate criterion: the repetition of common behaviour may merit technical review without increasing suspicion.

No incident needs priority review. 4 merit review and 5 are context observations. ICdetector recorded no alarm: all its signals are sub-threshold. 13 ICdetector records grouped into 9 episodes; 8 episodes checkable against the saved measurements.

- E2 · 25 Sept 12:02 · Review: LTE band change (B7->B8) in C1; once, sub-threshold.
- E3 · 25 Sept 12:33 · Review: LTE band change (B7->B20) in C2; once, sub-threshold.
- E8 · 29 Sept 14:05–14:26 · Review: LTE band change (B7->B8, B3->B8) in C1; 3 times, sub-threshold.
- E9 · 29 Sept 14:17–14:34 · Review: LTE band change (B7->B20, B3->B20) in C2; 2 times, sub-threshold.
- Context: LTE band change E1, E7; power deviation E4; new item at a usual place E5; alternating PCI in the same cell E6.
- Radio records degraded states with no nearby ICdetector signals (8 out of service); they do not form an incident.

With the criteria implemented, no observation can be attributed to a fake base station or another attack; nor can it be stated that there has been no problem at all.

2. Incidents

Table 1. Incidents in chronological order. «Observation» summarises what ICdetector recorded; «Evidence» is CellCase's checks

Ep.	Date	Observation (ICdetector)	Evidence (CellCase)	Interpretation	Technical interest	Spoofing
E1	25 Sept 06:23	LTE band change (B3->B20) in C3; once, sub-threshold	Change visible in the rows: B3->B20 (C12->C3); No recurrence on other days	Consistent with band management; not conclusive	Context	Insufficient evidence
E2	25 Sept 12:02	LTE band change (B7->B8) in C1; once, sub-threshold	Change visible in the rows: B7->B8 (C8->C1); Recurring: 2 days (E8)	Consistent with band management; not conclusive	Review	Insufficient evidence
E3	25 Sept 12:33	LTE band change (B7->B20) in C2; once, sub-threshold	Change visible in the rows: B7->B20 (C9->C2); Recurring: 2 days (E9)	Consistent with band management; not conclusive	Review	Insufficient evidence
E4	27 Sept 16:48	Power deviation (+19 dB) in C4; once, sub-threshold	Visible difference: -81 dBm against a median of -100 dBm (n=872, +19 dB); No recurrence on other days	Consistent with a change of position or band	Context	Insufficient evidence
E5	27 Sept 21:14–21:16	New item at a usual place in C5; once; also in: Stable-Site	No CellCase check for this pattern; No recurrence on other days	No specific interpretation	Context	Insufficient evidence
E6	29 Sept 13:53	Alternating PCI in the same cell in C6; once, sub-threshold	Visible: repeated PCI alternation; No recurrence on other days	Several possible causes; not distinguishable without neighbours	Context	More data needed
E7	29 Sept 14:00	LTE band change (B3->B20) in C7; once, sub-threshold	Change visible in the rows: B3->B20 (C10->C7); No recurrence on other days	Consistent with band management; not conclusive	Context	Insufficient evidence

Table 1. Incidents in chronological order. «Observation» summarises what ICdetector recorded; «Evidence» is …

Ep.	Date	Observation (ICdetector)	Evidence (CellCase)	Interpretation	Technical interest	Spoofing
E8	29 Sept 14:05–14:26	LTE band change (B7->B8, B3->B8) in C1; 3 times, sub-threshold	Change visible in the rows: B7->B8 (C9->C1), B3->B8 (C10->C1), B3->B8 (C11->C1); Recurring: 2 days (E2)	Consistent with band management; not conclusive	Review	Insufficient evidence
E9	29 Sept 14:17–14:34	LTE band change (B7->B20, B3->B20) in C2; 2 times, sub-threshold	Change visible in the rows: B7->B20 (C8->C2), B3->B20 (C10->C2); Recurring: 2 days (E3)	Consistent with band management; not conclusive	Review	Insufficient evidence

Radio, in all incidents: no degraded events recorded (the continuity of the Radio log is not demonstrated).

Group G1 (29 Sept 13:53–14:34): E6, E7, E8, E9 are close in time and shown together; each keeps its own data and conclusions. Closeness does not demonstrate a common cause.

Table 2. Short cell identifiers used in this report

Id.	Identity (MCC-MNC-TAC-CID-technology)	Id.	Identity
C1	`001-01-3002-1000545-LTE`	C7	`001-01-3002-1000061-LTE`
C2	`001-01-3002-1000543-LTE`	C8	`001-01-3002-1000544-LTE`
C3	`001-01-3001-1000069-LTE`	C9	`001-01-3002-1000542-LTE`
C4	`001-01-3001-1000070-LTE`	C10	`001-01-3002-1000060-LTE`
C5	`001-01-1099-9999001-LTE`	C11	`001-01-3003-1000989-LTE`
C6	`001-01-3003-1000953-LTE`	C12	`001-01-3001-1000980-LTE`

3. Incident details

E1 · 25 Sept 06:23 · LTE band change in C3 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 06:23:20 · history, C3: «[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)» — explanatory translation: “[sub-threshold] Forced band downgrade (1805MHz->791MHz, B3->B20)”

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B3->B20 (C12->C3).
- A single observation at 06:23:20: its duration is unknown.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- Verification in public databases for C3 (context): NOT_FOUND 56 rows.
- C3 is usual: 7 days as serving cell in the history.

What it means and what it does not allow us to conclude

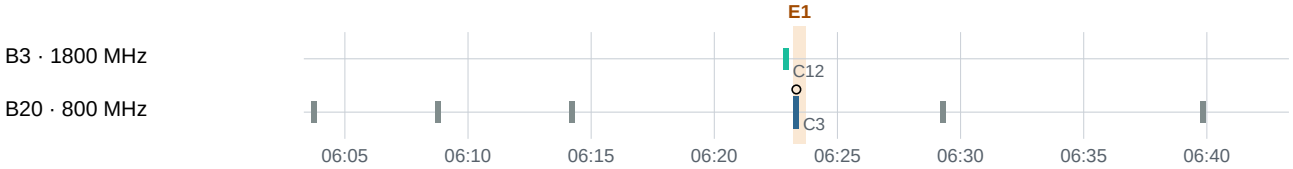
- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.

Figure 1. Band of the serving cell (E1)

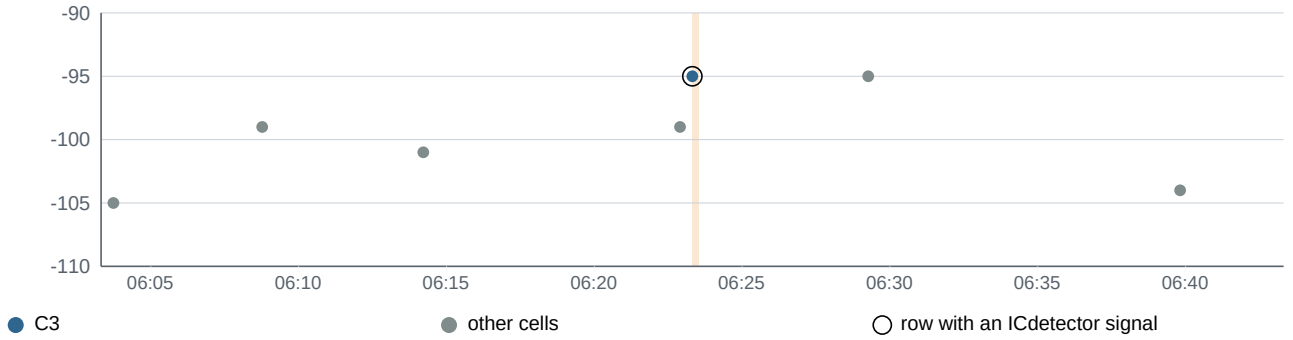
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 2. Serving cell power before and after (E1, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C4 (4 rows), C12 (1 row); during C3 (1 row); 30 min after C4 (4 rows). References: SYNTHETIC_patrones_historial.csv:1277.

E2 · 25 Sept 12:02 · LTE band change in C1 · Review

What was observed (recorded by ICdetector; verbatim quotes)

- 12:02:12 · history, C1: «[sub-umbral] Downgrade de banda forzado (2620MHz->925MHz, B7->B8)» — explanatory translation: “[sub-threshold] Forced band downgrade (2620MHz->925MHz, B7->B8)”

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B7->B8 (C8->C1).
- A single observation at 12:02:12: its duration is unknown.
- The same pattern appears in C1 on 2 different days (also in E8). This is recurrence across days, not continuous activity.
- Radio: no degraded events recorded.
- Verification in public databases for C1 (context): NOT_FOUND 4 rows.

What it means and what it does not allow us to conclude

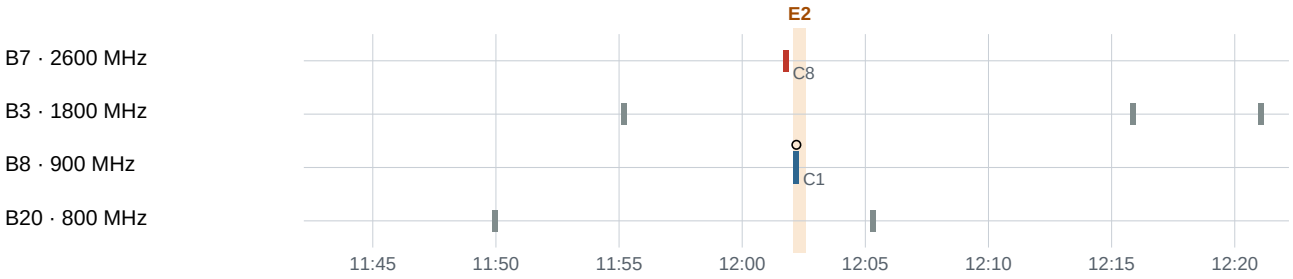
- Worth reviewing: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it recurs on 2 days; it is a single observation.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.
- Repeat the capture at the same place and time of day to see whether the pattern recurs.

Figure 3. Band of the serving cell (E2)

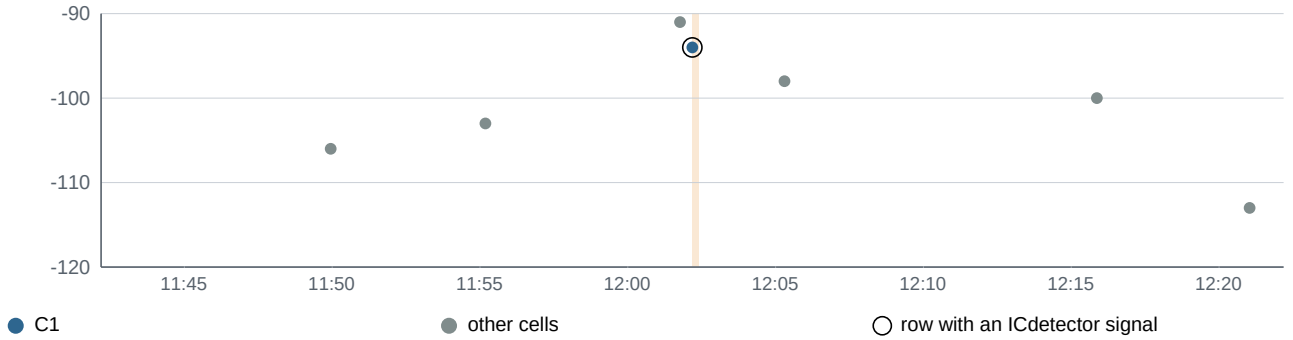
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 4. Serving cell power before and after (E2, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C10 (1 row), C4 (1 row), 001-01-3001-1000071-LTE (1 row); during C1 (1 row); 30 min after C4 (2 rows), 001-01-3001-1000071-LTE (1 row), C11 (1 row). References: SYNTHETIC_patrones_historial.csv:1234.

E3 · 25 Sept 12:33 · LTE band change in C2 · Review

What was observed (recorded by ICdetector; verbatim quotes)

- 12:33:07 · history, C2: «[sub-umbral] Downgrade de banda forzado (2620MHz->791MHz, B7->B20)» — explanatory translation: “[sub-threshold] Forced band downgrade (2620MHz->791MHz, B7->B20)”

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B7->B20 (C9->C2).
- A single observation at 12:33:07: its duration is unknown.
- The same pattern appears in C2 on 2 different days (also in E9). This is recurrence across days, not continuous activity.
- Radio: no degraded events recorded.
- Verification in public databases for C2 (context): VERIFIED 3 rows.

What it means and what it does not allow us to conclude

- Worth reviewing: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it recurs on 2 days; it is a single observation.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.
- Repeat the capture at the same place and time of day to see whether the pattern recurs.

Figure 5. Band of the serving cell (E3)

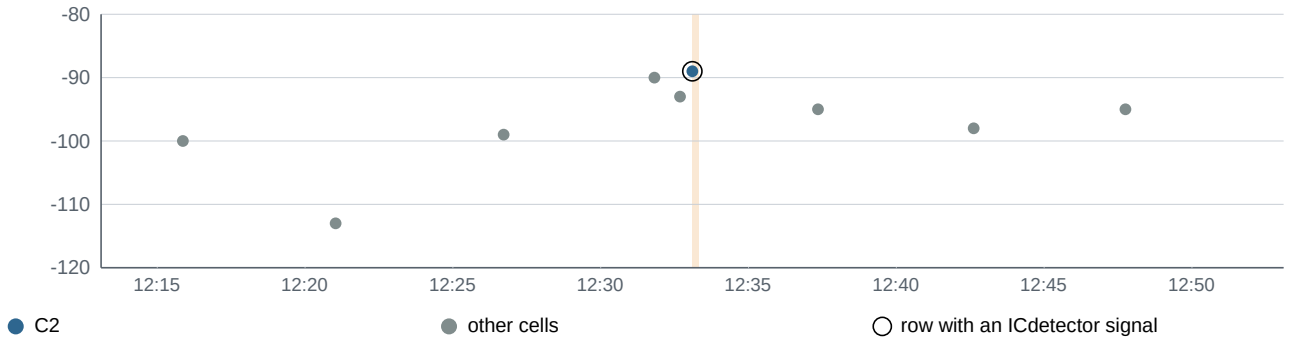
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 6. Serving cell power before and after (E3, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C4 (2 rows), 001-01-3001-1000071-LTE (1 row), C11 (1 row); during C2 (1 row); 30 min after C4 (2 rows), C9 (1 row). References: SYNTHETIC_patrones_historial.csv:1227.

E4 · 27 Sept 16:48 · Power deviation in C4 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 16:48:53 · history, C4: «[sub-umbral] Potencia anómala vs historial (+19dB)» — explanatory translation: “[sub-threshold] Anomalous power vs history (+19dB)”

What evidence supports it (CellCase checks)

- Against the reference for the same cell outside the episode: -81 dBm against a median of -100 dBm (n=872, +19 dB).
- A single observation at 16:48:53: its duration is unknown.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- Verification in public databases for C4 (context): NOT_FOUND 876 rows.
- C4 is usual: 9 days as serving cell in the history.

What it means and what it does not allow us to conclude

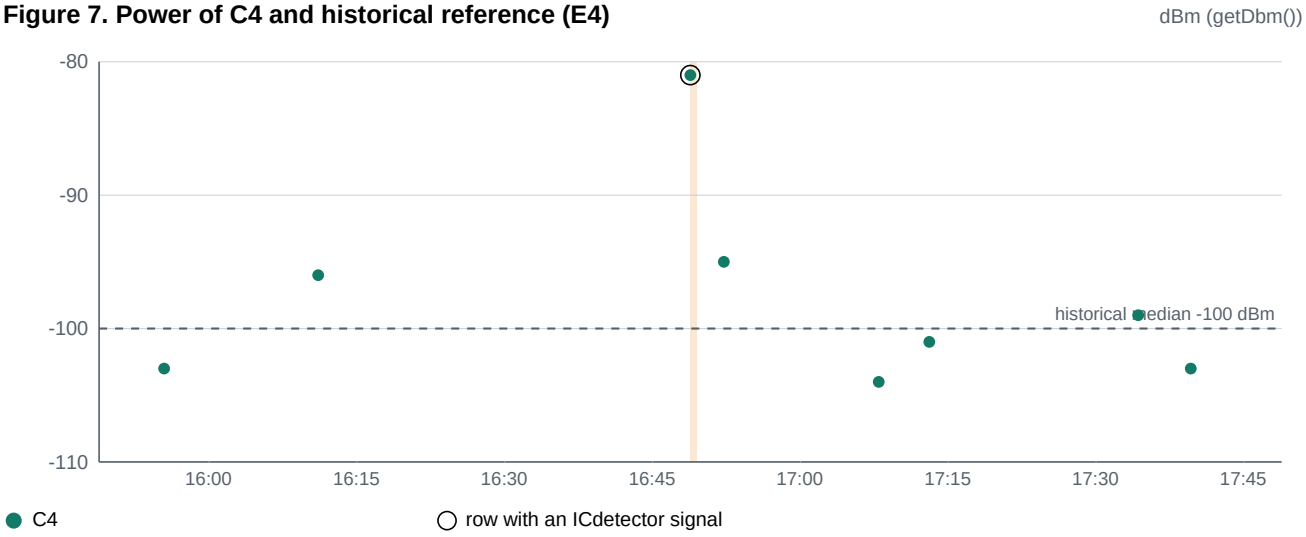
- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation.
- Power changes with the position relative to the station (indoors or outdoors, height, obstacles) and with the band in use.
- The difference cannot be attributed to another transmitter: the reference mixes different positions and times.

What to check next

- Compare with captures at the exact same spot (the reference mixes different positions).

SYNTHETIC DATA — NOT REAL

Figure 7. Power of C4 and historical reference (E4)



Rows of C4 within ±1 h. Dashed line: median of the same cell outside the episode (n=872 rows, mixing positions and times). No interpolation.

Serving cells: 30 min before C10 (1 row), C12 (1 row); during C4 (1 row); 30 min after C4 (3 rows), C3 (1 row). References: SYNTHETIC_patrones_historial.csv:855.

E5 · 27 Sept 21:14–21:16 · New item at a usual place in C5 · Context

What was observed (recorded by ICdetector; verbatim quotes)

- 21:14:00 · Stable-Site: «NEW_SERVING_WHILE_STATIC_AT_MATURE_SITE»

What evidence supports it (CellCase checks)

- CellCase has no check of its own on the measurements for this type of pattern.
- 1 observation between 21:14:00 and 21:16:30; brief repetition. The interval does not demonstrate continuous activity.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- C5 appears in 0 rows and on 0 days of the history (context; it does not change the priority).

What it means and what it does not allow us to conclude

- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); CellCase has no check implemented for this pattern (not assessable does not mean not contradictory); it is a single observation.
- ICdetector's heuristics are not calibrated: their activation alone has no specific interpretation.
- No specific cause can be concluded.

What to check next

- Check whether that cell keeps appearing at that place over the following days (the same follow-up the plain-language report proposes).

Serving cells: 30 min before C12 (2 rows), C4 (1 row); during —; 30 min after C4 (5 rows). References: SYNTHETIC_ICD-stable-site.zip!shadow_episodes.csv:2.

SYNTHETIC DATA — NOT REAL

E6 · 29 Sept 13:53 · Alternating PCI in the same cell in C6 · Context · G1

What was observed (recorded by ICdetector; verbatim quotes)

- 13:53:44 · history, C6: «[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)» — explanatory translation: “[sub-threshold] Unstable RF identity: same Cell ID alternating PCI on the same carrier (possible clone)”
- 13:54:36 · history, C6: «[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)» — explanatory translation: “[sub-threshold] Unstable RF identity: same Cell ID alternating PCI on the same carrier (possible clone)”

What evidence supports it (CellCase checks)

- Within the same full cell identity and carrier, the rows show repeated PCI alternation: C6, EARFCN 1300: PCI 101 (13:53:20) -> PCI 102 (13:53:44) -> PCI 101 (13:54:10) -> PCI 102 (13:54:36). This describes what the file records; it does not prove that each row is a new modem reading.
- A single observation at 13:53:44: its duration is unknown.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- Verification in public databases for C6 (context): NOT_FOUND 4 rows.
- C6 appears in 4 rows and on 1 day of the history (context; it does not change the priority).
- No forensic case covers the interval: neighbours and cycle-by-cycle diagnostics are unknown.

What it means and what it does not allow us to conclude

- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation.
- Several carriers or sectors can share part of the identity, and the modem may report data from the previous cell during a change.
- It may also be due to PCI re-planning by the operator.
- The alternation is visible in the data; without neighbours and diagnostics (forensic case) these explanations cannot be told apart.
- Those causes cannot be told apart without the neighbours and diagnostics of a forensic case.

What to check next

- If ICdetector opened a forensic case in that interval, export it: it includes neighbours and cycle-by-cycle diagnostics.
- Check whether the same PCI and EARFCN appear with another identity in other captures.

Figure 8. PCI and EARFCN of C6 (E6)

local time in Europe/Madrid (declared assumption)



History rows with the same full cell identity (MCC, MNC, area, Cell ID and technology) within ±30 min; one lane per PCI and carrier combination. Several lanes on the same carrier indicate PCI changes; the check table says whether it is a single PCI change, a PCI return or a repeated PCI alternation. Dotted mark: row with an ICdetector signal.

Serving cells: 30 min before C8 (9 rows), C11 (9 rows), C10 (6 rows); during C6 (3 rows), C10 (1 row), C8 (1 row); 30 min after C9 (16 rows), C10 (9 rows), C8 (6 rows). References: SYNTHETIC_patrones_historial.csv:500; SYNTHETIC_patrones_historial.csv:496.

SYNTHETIC DATA — NOT REAL

E7 · 29 Sept 14:00 · LTE band change in C7 · Context · G1

What was observed (recorded by ICdetector; verbatim quotes)

- 14:00:42 · history, C7: «[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)» — explanatory translation: “[sub-threshold] Forced band downgrade (1805MHz->791MHz, B3->B20)”

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B3->B20 (C10->C7).
- A single observation at 14:00:42: its duration is unknown.
- It does not recur on other days with the same cell.
- Radio: no degraded events recorded.
- Verification in public databases for C7 (context): NOT_FOUND 1 row.
- C7 appears in 1 row and on 1 day of the history (context; it does not change the priority).

What it means and what it does not allow us to conclude

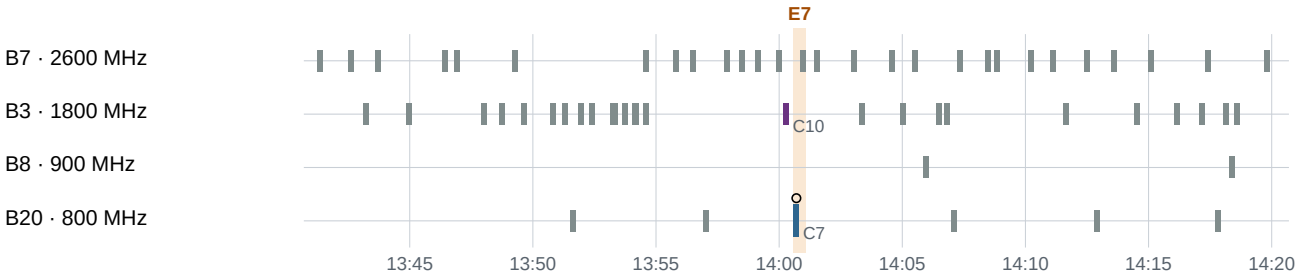
- Context observation: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it is a single observation.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.

Figure 9. Band of the serving cell (E7)

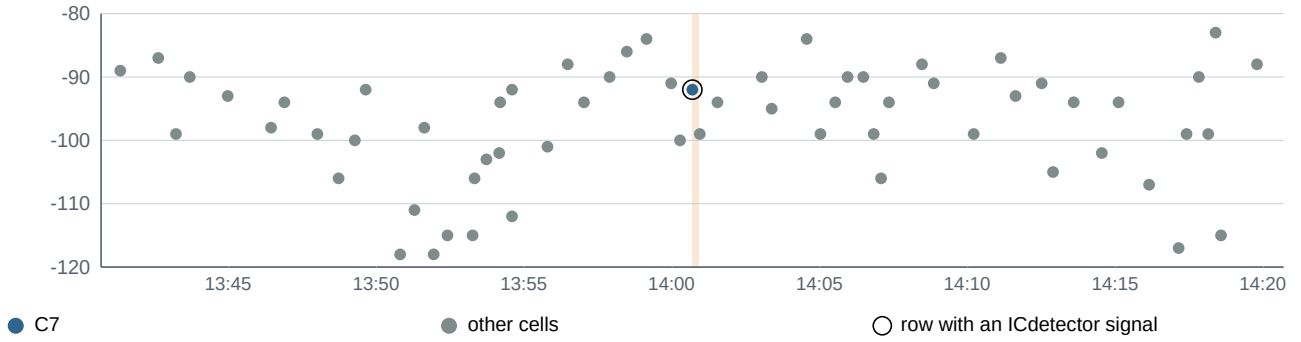
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 10. Serving cell power before and after (E7, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C9 (10 rows), C11 (9 rows), C8 (9 rows); during C7 (1 row); 30 min after C9 (11 rows), C10 (10 rows), C8 (8 rows). References: SYNTHETIC_patrones_historial.csv:487.

SYNTHETIC DATA — NOT REAL

E8 · 29 Sept 14:05–14:26 · LTE band change in C1 · Review · G1

What was observed (recorded by ICdetector; verbatim quotes)

- 14:05:57 · history, C1: «[sub-umbral] Downgrade de banda forzado (2620MHz->925MHz, B7->B8)» — explanatory translation: “[sub-threshold] Forced band downgrade (2620MHz->925MHz, B7->B8)”
- 14:18:24 · history, C1: «[sub-umbral] Downgrade de banda forzado (1805MHz->925MHz, B3->B8)» — explanatory translation: “[sub-threshold] Forced band downgrade (1805MHz->925MHz, B3->B8)”
- 14:26:58 · history, C1: «[sub-umbral] Downgrade de banda forzado (1805MHz->925MHz, B3->B8)» — explanatory translation: “[sub-threshold] Forced band downgrade (1805MHz->925MHz, B3->B8)”

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B7->B8 (C9->C1), B3->B8 (C10->C1), B3->B8 (C11->C1).
- 3 observations between 14:05:57 and 14:26:58. It recurs within the episode, but the interval between samples does not demonstrate continuous activity.
- The same pattern appears in C1 on 2 different days (also in E2). This is recurrence across days, not continuous activity.
- Radio: no degraded events recorded.
- Verification in public databases for C1 (context): NOT_FOUND 4 rows.

What it means and what it does not allow us to conclude

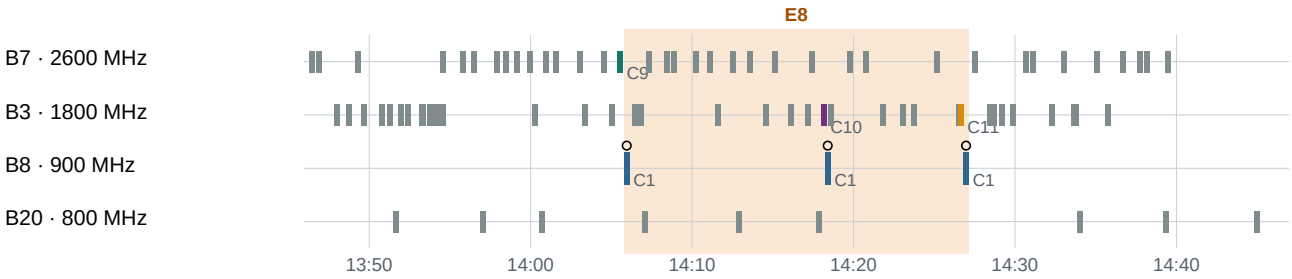
- Worth reviewing: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it recurs on 2 days; it recurs within the episode (3 observations).
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- Recurrence in the same cell may reflect its configuration or where the phone was.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.
- Repeat the capture at the same place and time of day to see whether the pattern recurs.

Figure 11. Band of the serving cell (E8)

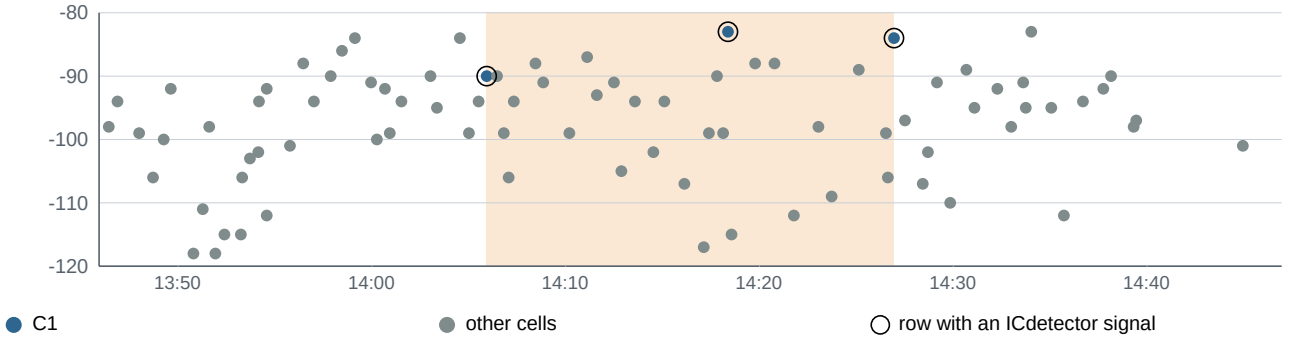
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 12. Serving cell power before and after (E8, LTE)

dBm (RSRP; magnitude of getDbm())



Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C9 (13 rows), C10 (9 rows), C8 (8 rows); during C10 (7 rows), C9 (7 rows), C11 (6 rows); 30 min after C8 (7 rows), C10 (4 rows), C11 (3 rows). References: SYNTHETIC_patrones_historial.csv:479; SYNTHETIC_patrones_historial.csv:459; SYNTHETIC_patrones_historial.csv:449.

E9 · 29 Sept 14:17–14:34 · LTE band change in C2 · Review · G1

What was observed (recorded by ICdetector; verbatim quotes)

- 14:17:50 · history, C2: «[sub-umbral] Downgrade de banda forzado (2620MHz->791MHz, B7->B20)» — explanatory translation: “[sub-threshold] Forced band downgrade (2620MHz->791MHz, B7->B20)”
- 14:34:03 · history, C2: «[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)» — explanatory translation: “[sub-threshold] Forced band downgrade (1805MHz->791MHz, B3->B20)”

What evidence supports it (CellCase checks)

- In the history, the row before the signal shows the band change: B7->B20 (C8->C2), B3->B20 (C10->C2).
- 2 observations between 14:17:50 and 14:34:03; brief repetition. The interval does not demonstrate continuous activity.
- The same pattern appears in C2 on 2 different days (also in E3). This is recurrence across days, not continuous activity.
- Radio: no degraded events recorded.
- Verification in public databases for C2 (context): VERIFIED 3 rows.

What it means and what it does not allow us to conclude

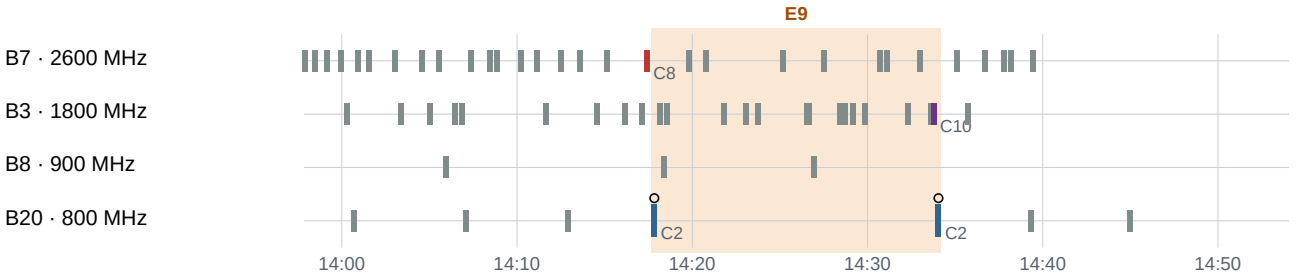
- Worth reviewing: ICdetector did not record it as an alarm (it did not reach its threshold); the behaviour is confirmed in the measurements; it recurs on 2 days.
- A band change within LTE is common in network management: load balancing, carrier aggregation or coverage; low bands (800/900 MHz) are used to cover indoors and cell edges.
- Recurrence in the same cell may reflect its configuration or where the phone was.
- It does not show what caused the band change or whether the network or the phone decided it.

What to check next

- Compare the band used at the same place and time of day on other days.
- Repeat the capture at the same place and time of day to see whether the pattern recurs.

Figure 13. Band of the serving cell (E9)

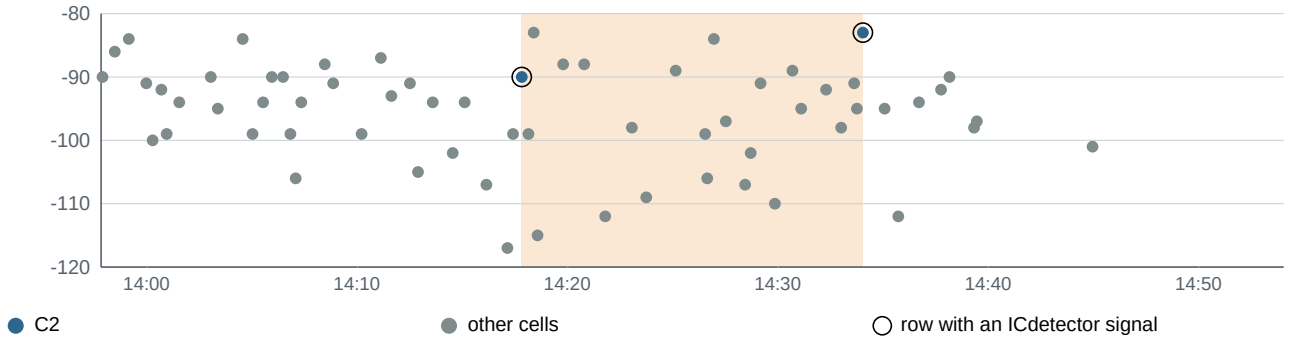
local time in Europe/Madrid (declared assumption)



Each mark is a history row in the serving cell's band (Bands or EARFCN column). Dotted mark: row with an ICdetector signal. Labels and colours: rows with a signal and the row before each one; the rest in grey. Nothing is drawn between rows. Source: history.

Figure 14. Serving cell power before and after (E9, LTE)

dBm (RSRP; magnitude of getDbm())



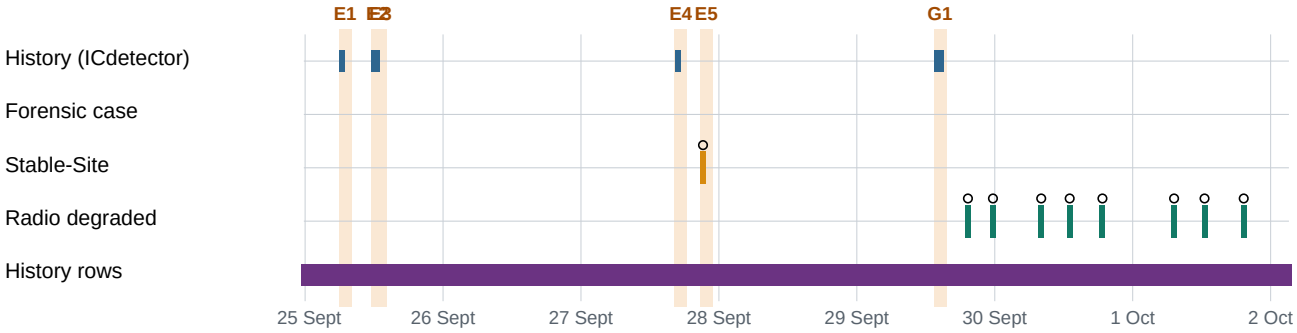
Measured points, neither joined nor interpolated; the bands mark from the first to the last record of each episode, not continuous activity. Different bands have different coverage: power is not directly comparable between bands.

Serving cells: 30 min before C9 (14 rows), C10 (10 rows), C11 (8 rows); during C10 (7 rows), C11 (6 rows), C8 (4 rows); 30 min after C8 (4 rows), C4 (2 rows), C9 (1 row). References: SYNTHETIC_patrones_historial.csv:461; SYNTHETIC_patrones_historial.csv:437.

4. Chronology

Figure 15. Signals by source over the period of the data

local time in Europe/Madrid (declared assumption)

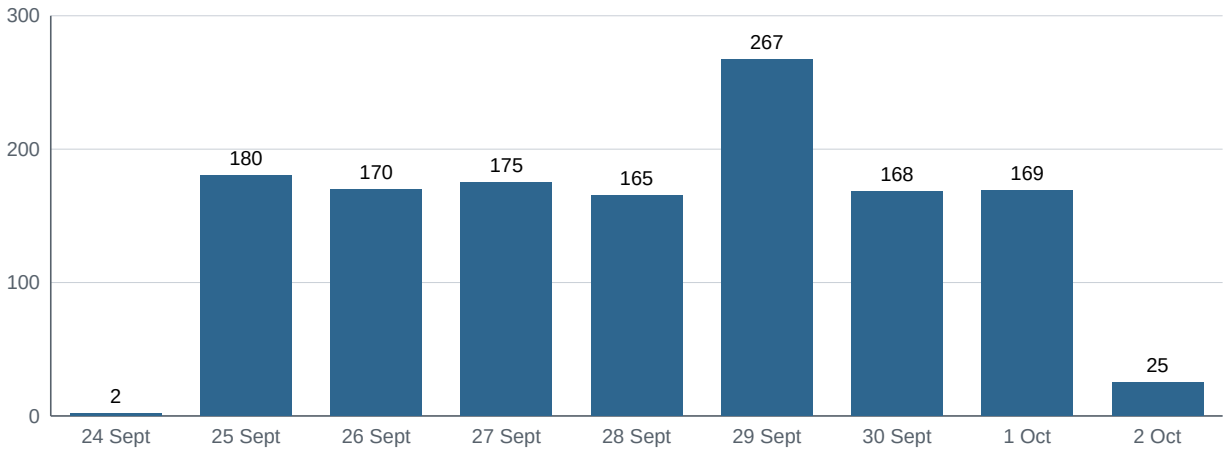


Each mark is a record in its source (tall marks: not sub-threshold). «History rows» shows when there is sampling; the gaps are intervals with no rows. The bands are episodes or groups of episodes (G).

SYNTHETIC DATA — NOT REAL

Figure 16. History rows by local date

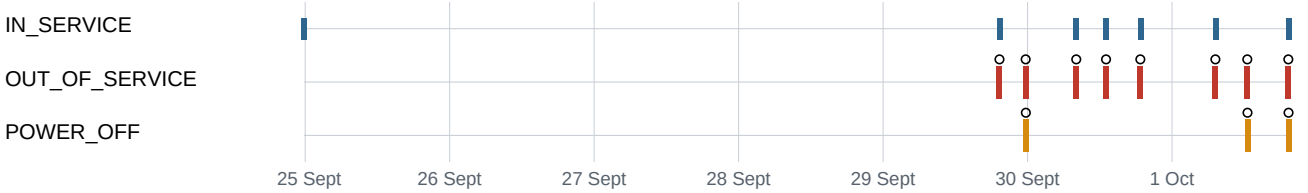
Rows per day



The phone's local date; the height reflects ICdetector's sampling, not the phone's activity.

Figure 17. Service state changes recorded by Radio (whole period)

local time in Europe/Madrid (declared assumption)



Each mark is a recorded state change. Between marks it is unknown whether logging was continuous: the intervals are not drawn as sustained states. Source: Radio export (UTC).

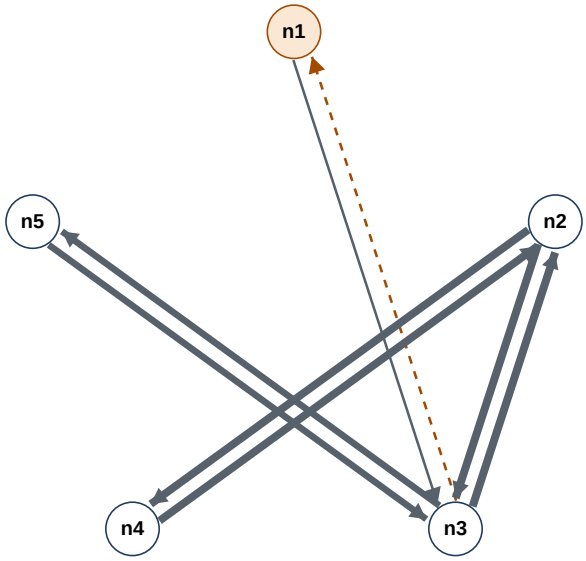
5. Network context

Topology, Geometry and Stable-Site are cumulative aggregates: they describe usual routes and places, not individual events, and they are not added to the history observations. Radio and forensic cases do have exact times.

Transitions between cells (Topology)

Content	8 directed routes, 192 cumulative handovers, generated 2026-09-28 22:00:00
Limit	Below the limit of 250 routes; this does not prove that it covers the whole campaign (retention, partial files).
Last state	1 route with the last handover marked FAILED by the collector (it describes only the last one, not all).

Figure 18. Transitions observed between cells (Topology aggregate)



n1 = 001-01-1099-9999001-LTE · 2 handovers · in an episode
n2 = 001-01-1001-1000002-LTE · 154 handovers
n3 = 001-01-1001-1000001-LTE · 119 handovers
n4 = 001-01-1002-1000003-LTE · 73 handovers
n5 = 001-01-1001-5000000001-NR · 36 handovers
Only routes present in transitions.csv are drawn; the thickness grows with the cumulative number of handovers (logarithmic scale). Dashed line: the last handover on that route was FAILED. Highlighted nodes: cells that appear in some episode. Selection: cells from episodes (up to 6) and then those with the most handovers, up to 14. Omitted: 0 cells and 0 routes. The aggregate does not say when each handover happened.

Service state (Radio)

Content	18 state changes between 2026-09-24 23:46:33 and 2026-10-01 19:25:00
States	OUT_OF_SERVICE 8; IN_SERVICE 7; POWER_OFF 3
Operator different from the SIM without roaming	0
Degraded states with no other signals	8 out of service (they do not form an episode)
Consistency with the history	See «Agreements and differences between sources» (X-SVC).

Usual places (Stable-Site)

Each place is recorded under up to four overlapping site keys, so there is no single total of observations: they are shown per key. The keys are hashed summaries of the position; CellCase does not turn them into coordinates.

Table 3. Site keys with the most observations (at most 10 of 8)

Key	State	Serving days	Static days	Serving obs.	Cells
3828ccfb43…	ACTIVE	56	54	2,140	3
59634f59dc…	ACTIVE	56	54	2,140	3
a7cd027657…	ACTIVE	56	54	2,140	3
94f9343de8…	ACTIVE	56	54	2,140	3
e2d6ca8cfd…	SHADOW_READY	6	6	610	1
73f97a25ef…	SHADOW_READY	6	6	610	1
423f4bfdd2…	SHADOW_READY	6	6	610	1
941649a3f1…	SHADOW_READY	6	6	610	1

Shadow episodes: 2, of which 1 were recorded as a possible novelty (collector result; in 2.10.5 they do not block anything).

Mobility (Geometry)

Content 5 cells, 8 routes and 4 trips; no coordinates

Familiarity KNOWN_ON_ROUTE 3; UNKNOWN_ON_ROUTE 2. KNOWN_ON_ROUTE means a known route, neither trusted nor verified.

Trips 1 open; 1 closed that do not meet the validity rule (discarded by ICdetector).

Limit Below the limit of 1,000; it does not prove complete coverage.

6. Agreements and differences between sources

Compares the sources of this case with each other, using full cell identities and times with a single instant. Each result is classified as an agreement, a difference, complementary information or not comparable. An absence from a file does not prove a contradiction, and a difference does not prove a network anomaly or an IMSI catcher. Almost all sources come from ICdetector: an agreement between them checks that the files are consistent; it is not an independent confirmation. The cross-checks are context and do not change the assessment.

Results: 5 agreements, 1 difference, 3 complementary and 7 not comparable. The totals count check results (a row, a cell, a route or a Stable-Site episode compared), not incidents or independent evidence.

Summary by check

Table 4. Checks cross/4: 14 groups covering 16 results; each result is in a single group. Examples: the first results of each group, with their reference (file, entry and physical line).

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-SVC Service state History / Radio	Agreement 1 result · 1,151 items compared · 24 Sept 23:46 – 1 Oct 12:28	Both sources record the same value at the compared moments. • 1,151 rows. state: History IN_SERVICE · Radio IN_SERVICE (1,151). 24 Sept 23:46 – 1 Oct 12:28. Ref.: SYNTHETIC_patrones_historial.csv:1322; SYNTHETIC_patrones_radio.csv:2	Both sources read Android's service state; Radio only records changes and its continuity is not demonstrated.
X-SVC Service state History / Radio	Difference 1 result · 111 items compared · 29 Sept 23:48 – 1 Oct 19:22	The sources record different values for the same entity at comparable moments. • 111 rows. state: History IN_SERVICE · Radio POWER_OFF (111). 29 Sept 23:48 – 1 Oct 19:22. window ±1 min. Ref.: SYNTHETIC_patrones_historial.csv:364; SYNTHETIC_patrones_historial.csv:365	Both sources read Android's service state; Radio only records changes and its continuity is not demonstrated. A difference does not prove a network anomaly or an IMSI catcher. Next step: review the cited rows in the original files.

Table 4. Checks cross/4: 14 groups covering 16 results; each result is in a single group. Examples: the first res…

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-SVC Service state History / Radio	Not comparable 1 result · 57 items compared · 1 Oct 19:28 – 2 Oct 03:11	Rows before the first Radio event or after the last one: Radio does not say until when it covers. • 57 rows. 1 Oct 19:28 – 2 Oct 03:11. Ref.: SYNTHETIC_patrones_historial.csv:56; SYNTHETIC_patrones_historial.csv:57	Both sources read Android's service state; Radio only records changes and its continuity is not demonstrated. Next step: provide a Radio export that covers that period.
X-SVC Service state History / Radio	Not comparable 1 result · 2 items compared · 29 Sept 19:21 – 30 Sept 18:45	Moments a few seconds from a change: the history time is the insertion time with a resolution of seconds, so it cannot be paired reliably. • 2 rows. state: History IN_SERVICE · Radio OUT_OF_SERVICE (2). 29 Sept 19:21 – 30 Sept 18:45. window ±1 min. Ref.: SYNTHETIC_patrones_historial.csv:233; SYNTHETIC_patrones_historial.csv:394	Both sources read Android's service state; Radio only records changes and its continuity is not demonstrated.
X-CELL Serving cells History / Topology, Geometry	Complementary 1 result · 5 items compared · 5 entities	Cells that only appear in the aggregate: it is cumulative and may cover periods that the history does not include. • 5 cells: C13, C14 and 3 more. C13: handovers: Topology 119. and 7 more. Ref.: SYNTHETIC_ICD-geometry.zip! geometry_cells.csv:2; SYNTHETIC_ICD-geometry.zip! geometry_cells.csv:3	Both sources come from ICdetector.
X-CELL Serving cells History / Topology, Geometry, Stable-Site	Complementary 1 result · 12 items compared · 12 entities	History cells that the aggregate does not contain: the aggregate only records part of the data (recent routes, places with a precise position) and may be from another time. • 12 cells: C3, C4 and 10 more. C3: rows: History 56. and 7 more. Ref.: SYNTHETIC_patrones_historial.csv:2; SYNTHETIC_patrones_historial.csv:49	Both sources come from ICdetector.
X-CELL Serving cells History / Stable-Site	Complementary 1 result · 4 items compared · 4 entities	Cells that only appear in the aggregate: it is cumulative and may cover periods that the history does not include. • 4 cells: C13, C14 and 2 more. C13: days as serving cell (max. per key): Stable-Site 56. and 3 more. Ref.: SYNTHETIC_ICD-stable-site.zip! site_cells.csv:2; SYNTHETIC_ICD-stable-site.zip! site_cells.csv:3	Both sources come from ICdetector.
X-ROUTE Cell changes and routes History / Topology, Geometry	Not comparable 1 result · 23 items compared · 23 entities · 29 Sept 10:06 – 1 Oct 13:21	Changes after the aggregate was exported: it could not contain them. • 23 routes: C3 -> C9, C3 -> C8 and 21 more. C3 -> C9: changes: History 2. and 7 more. 29 Sept 10:06 – 1 Oct 13:21. window ±10 min. Ref.: SYNTHETIC_patrones_historial.csv:268; SYNTHETIC_patrones_historial.csv:269	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation. Next step: export the aggregate again after those dates.
X-ROUTE Cell changes and routes History / Topology, Geometry	Not comparable 1 result · 44 items compared · 44 entities · 25 Sept 06:22 – 1 Oct 20:36	The aggregate does not contain those cells: there are no routes to compare with. • 44 routes: C3 -> C4, C3 -> C12 and 42 more. C3 -> C4: changes: History 14. and 7 more. 25 Sept 06:22 – 1 Oct 20:36. window ±10 min. Ref.: SYNTHETIC_patrones_historial.csv:666; SYNTHETIC_patrones_historial.csv:667	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation.

Table 4. Checks cross/4: 14 groups covering 16 results; each result is in a single group. Examples: the first res…

Comparison	Result and scope	Explanation and examples	Limitations and next step
X-ROUTE Cell changes and routes History / Topology, Geometry	Not comparable 1 result · 1 item compared · 2 entities · 29 Sept 13:54	History rows with different identities at the same instant: it is not known which was the serving cell or in what order, so no changes are derived to, from or through that instant, and they are not paired with another source. • 1 instant: C8, C6. 29 Sept 13:54: serving cell: History C8 + C6. 29 Sept 13:54. Ref.: SYNTHETIC_patrones_historial.csv:496; SYNTHETIC_patrones_historial.csv:497	Both sources come from ICdetector.
X-AGG Route counters Topology / Geometry	Agreement 1 result · 7 items compared · 7 entities · 28 Sept 22:00 – 29 Sept 09:00	Both sources record the same value at the compared moments. • 7 routes: C13 -> C15, C13 -> C5 and 5 more. C13 -> C15: handovers: Topology 18 · Geometry 18. and 7 more. 28 Sept 22:00 – 29 Sept 09:00. Ref.: SYNTHETIC_ICD-geometry.zip! geometry_edges.csv:3; SYNTHETIC_ICD-geometry.zip! geometry_edges.csv:4	Counters from the same ICdetector database at different times.
X-AGG Route counters Topology / Geometry	Agreement 1 result · 1 item compared · 1 entity · 28 Sept 22:00 – 29 Sept 09:00	Compatible counters: the later export has equal or higher values (aggregates are updated between exports). • 1 route: C13 -> C14. C13 -> C14: handovers: Topology 41 · Geometry 42. and 1 more. 28 Sept 22:00 – 29 Sept 09:00. Ref.: SYNTHETIC_ICD-geometry.zip! geometry_edges.csv:2; SYNTHETIC_ICD-topology.zip! transitions.csv:2	Counters from the same ICdetector database at different times.
X-SHADOW Stable-Site episode Stable-Site / History	Not comparable 2 results · 0 items compared · 4 entities · 27 Sept 21:14 – 28 Sept 08:26	There are no history rows in the window: the history is sampled. • 0 rows in the window: C5, 5e0a7c1d…. 27 Sept 21:14 – 27 Sept 21:16. window ±5 s. • 0 rows in the window: C16, 5e0a7c1d…. 28 Sept 08:25 – 28 Sept 08:26. window ±5 s. Ref.: SYNTHETIC_ICD-stable-site.zip! shadow_episodes.csv:2; SYNTHETIC_patrones_historial.csv:821; SYNTHETIC_ICD-stable-site.zip! shadow_episodes.csv:3	Both sources come from ICdetector. Next step: if ICdetector opened a forensic case at that time, export it.
X-SHADOW Stable-Site episode Stable-Site / Topology	Agreement 2 results · 2 items compared · 4 entities · 27 Sept 21:14 – 28 Sept 08:26	The route recorded its last change within the window of the Stable-Site episode (margin ±1 min). It is compatible with the same cell change, but coinciding within the margin does not prove that it is exactly the same event. • 1 episode: C13 -> C5, 5e0a7c1d…. handovers: Topology 1. and 2 more. 27 Sept 21:14 – 27 Sept 21:16. window ±1 min. • 1 episode: C14 -> C16, 5e0a7c1d…. handovers: Topology 37. and 2 more. 28 Sept 08:25 – 28 Sept 08:26. window ±1 min. Ref.: SYNTHETIC_ICD-stable-site.zip! shadow_episodes.csv:2; SYNTHETIC_ICD-topology.zip! transitions.csv:4; SYNTHETIC_ICD-stable-site.zip! shadow_episodes.csv:3	Both sources note cell changes seen by ICdetector: even if it is the same change, it would not be an independent confirmation.

Differences and cross-checks linked to the episodes

Table 5. Differences and results linked to episodes, grouped by episode and type so that the same context is not repeated. Their limitations are in the summary.

Episodes	Comparison	Result	Details and references
E5	X-SHADOW Stable-Site episode Stable-Site / History	Not comparable	There are no history rows in the window: the history is sampled. 0 rows in the window: C5, 5e0a7c1d…. 27 Sept 21:14 – 27 Sept 21:16. window ±5 s. Ref.: SYNTHETIC_ICD-stable-site.zip! shadow_episodes.csv:2; SYNTHETIC_patrones_historial.csv:821; SYNTHETIC_patrones_historial.csv:822
E5	X-SHADOW Stable-Site episode Stable-Site / Topology	Agreement	The route recorded its last change within the window of the Stable-Site episode (margin ±1 min). It is compatible with the same cell change, but coinciding within the margin does not prove that it is exactly the same event. 1 episode: C13 -> C5, 5e0a7c1d…. handovers: Topology 1. time: Stable-Site 27 Sept 21:14 · Topology 27 Sept 21:14. route: Stable-Site C13 -> C5 · Topology C13 -> C5. 27 Sept 21:14 – 27 Sept 21:16. window ±1 min. Ref.: SYNTHETIC_ICD-stable-site.zip! shadow_episodes.csv:2; SYNTHETIC_ICD-topology.zip! transitions.csv:4
E6	X-ROUTE Cell changes and routes History / Topology, Geometry	Not comparable	History rows with different identities at the same instant: it is not known which was the serving cell or in what order, so no changes are derived to, from or through that instant, and they are not paired with another source. 1 instant: C8, C6. 29 Sept 13:54: serving cell: History C8 + C6. 29 Sept 13:54. Ref.: SYNTHETIC_patrones_historial.csv:496; SYNTHETIC_patrones_historial.csv:497
—	X-SVC Service state History / Radio	Difference	The sources record different values for the same entity at comparable moments. 111 rows. state: History IN_SERVICE · Radio POWER_OFF (111). 29 Sept 23:48 – 1 Oct 19:22. window ±1 min. Ref.: SYNTHETIC_patrones_historial.csv:364; SYNTHETIC_patrones_historial.csv:365; SYNTHETIC_patrones_historial.csv:366 …

The full list of the 16 results is not included: it is an optional annex, off by default. The summary totals are the same.

New short identifiers in this table: C13 = 001-01-1001-1000001-LTE; C14 = 001-01-1001-1000002-LTE; C15 = 001-01-1001-5000000001-NR; C16 = 001-01-1002-1000003-LTE; C17 = 001-01-3001-1000071-LTE.

7. Data sources and quality

Declared origin: synthetic test data. Automatic signs of test data in 5 files (SYNTHETIC_patrones_historial.csv, SYNTHETIC_patrones_radio.csv, SYNTHETIC_ICD-topology.zip, SYNTHETIC_ICD-stable-site.zip, SYNTHETIC_ICD-geometry.zip): the file name indicates synthetic data; MCC 001 (test network); operator 00101 (test network). These signs do not decide the origin.

Table 6. Files provided

File	Type	State	Content	Period
SYNTHETIC_patrones_historial.csv	Observation history	Imported	1,321 rows	2026-09-24 23:46 – 2026-10-02 03:11 (phone's local time)
SYNTHETIC_patrones_radio.csv	Radio: service state events	Imported	18 events	24 Sept 23:46 – 1 Oct 19:25
SYNTHETIC_ICD-topology.zip	Topology	Imported	8 routes · 192 handovers (aggregate)	Routes last seen between 27 Sept 18:02 and 28 Sept 08:25 (this is not coverage)
SYNTHETIC_ICD-stable-site.zip	Stable-Site	Imported	8 site keys · 2 episodes	First and last aggregated record: 3 Aug 20:00 – 28 Sept 23:55
SYNTHETIC_ICD-geometry.zip	Geometry	Imported	8 routes · 4 trips	Trips between 27 Sept 21:13 and 29 Sept 08:00

Internal integrity: SYNTHETIC_ICD-topology.zip: SHA256SUMS verified (4 entries). SHA256SUMS only detects changes made after the export; it does not prove authorship, date or chain of custody.

History: quality and time interpretation

Rows	1,321 read; 0 discarded because of their structure; 12 complete serving identities
Valid fields	GNSS 90.8% · PCI 100.0% · ARFCN 100.0% · RSRQ 100.0% · SINR 100.0% · service state 100.0%
Discarded values	0 sentinels («N/A», 2147483647…) and 0 out of range, treated as missing; the original is kept
Timing Advance	STUB_ZERO 1,321. Only LTE_INDEX and GSM_INDEX can be converted to distance; STUB_ZERO indicates a modem that constantly reports 0.
Times	1,321 with a unique UTC instant; 0 ambiguous and 0 non-existent because of a daylight saving change (not resolved)
Sampling	median between rows 5 min 39 s; 0 intervals of 60 min or more

Analyses not available

Table 7. Analyses not available and reason

Analysis	State	Reason
Fine-grained incident chronology (forensic case)	Not provided	The required export has not been provided.

Not provided: the export is missing. Not compatible: it was provided but does not have the expected structure. Import failed: the file is damaged, unreadable or was rejected for safety. Not assessable: there are data, but not enough for that analysis.

8. Methodology

Each history row with an ICdetector reason, each forensic case and each episode recorded by Stable-Site is a signal. Signals from the same cell and of the same pattern type within <= 30 min form an episode; episodes separated by <= 15 min are presented in a group, without merging them. Radio is associated with an episode if a degraded state falls within <= 5 min.

- Observation: what ICdetector recorded, quoted verbatim.
- CellCase check: comparison with the history measurements (band by EARFCN in the previous row; PCI per full cell identity and carrier within ±30 min; power against the median of the same cell outside the episode, with at least 5 rows and a difference >= 10 dB).
- Persistence (within the episode): >= 3 distinct moments over >= 10 min; signals within <= 60 s count as one moment. Recurrence: same cell and pattern on >= 2 days.
- Priority review: an ICdetector alarm (a collector warning or a forensic case opened by an alarm), complete identities, behaviour confirmed in the measurements («not contradictory» is not enough) and, in addition, persistence or recurrence.
- Review: an ICdetector alarm; or behaviour confirmed in the measurements with recurrence or persistence. An observation forensic case is not an alarm, and a coincidence with Radio is context, not independent evidence: neither raises the priority.
- Context: everything else. Verification statuses and how usual a cell is are reported, but they do not change the priority.
- Versions: readers/2.4.1, cellcase-rules/3, episodes/7, cross/4 (changes in docs/episode-rules.md and docs/cross-sources.md).

9. Conclusion and next steps

Conclusion

- No incident needs priority review. Worth reviewing: E2, E3, E8, E9.
- ICdetector recorded no alarms: sub-threshold signals are only raised to review when the behaviour is visible in the measurements and recurs.
- Assessment for the spoofing question: Observations that need more data.
- No attribution to fake base stations or other attacks can be made: CellCase has no criteria implemented and evaluated for that. The lack of indications does not prove that there has been no problem at all either.

Next steps

- E2, E3, E8, E9: compare the band used at the same place and time of day on other days.
- If ICdetector opened forensic cases on those dates, export them and add them to the case.

10. Limitations

- ICdetector's heuristics are not calibrated; «sub-threshold» means it did not reach its alarm threshold. Its texts are quoted as written, in the original Spanish (e.g. «forzado», "forced"), and they are not CellCase conclusions.
- History, forensic case and Stable-Site come from the same collector: they are not counted as independent evidence. Radio is the only source from another family; a coincidence with Radio is reported as context, not as independent evidence, and the continuity of its log is not demonstrated.
- A single observation has an unknown duration; an interval between observations does not demonstrate continuous activity.
- Verification statuses (VERIFIED, NOT_FOUND…) are context from incomplete public databases: they neither authenticate nor rule out a base station and do not change the priority.
- Topology, Geometry and Stable-Site are aggregates with no time per event: they describe what is usual, they do not reconstruct specific events.
- Analyses not available: fine-grained incident chronology (forensic case) (not provided).
- The device, the time zone and the origin of the data are statements by the user; no discrepancies with the files have been detected.

11. Analyst's notes

The analyst has not added notes.

Annexes

A. Files, entries and integrity

Table 8. Fingerprint of each file provided

File	File SHA-256	Size	Internal integrity
SYNTHETIC_patrones_historial.csv	`ff2a93d0454e955e8be54a6c61c8482794974c6dce9cd3e2cf6be7f0327d7178`	223.7 KiB	The format does not include SHA256SUMS
SYNTHETIC_patrones_radio.csv	`dbb7a40acecfa34a2caa7eb808ae7b4237d88b97dbfbdea776c6f9f2775bfb96`	1.9 KiB	The format does not include SHA256SUMS
SYNTHETIC_ICD-topology.zip	`f2bcdd7236d1273f513e253d74812f694f88b8290d355681a8500c172dd9e7c5`	2.0 KiB	SHA256SUMS verified (4 entries)
SYNTHETIC_ICD-stable-site.zip	`bb7717eecbb6d5260f303a611b27e9a58f5a2dc10e19ae9f3ee0943726bf8001`	2.1 KiB	The format does not include SHA256SUMS
SYNTHETIC_ICD-geometry.zip	`4398c10250a06febb5142920fedf367065a8ffaa8dabd9621b6049b30a1b6455`	2.5 KiB	The format does not include SHA256SUMS

SHA-256 computed by CellCase over the bytes of the file when it was imported. It allows checking that the stored copy has not changed; it does not prove authenticity, authorship, capture date or chain of custody.

Table 9. Internal ZIP entries

File	Entry	Role	Rows	Size
SYNTHETIC_ICD-topology.zip	cells.csv	Check	5	381 B
SYNTHETIC_ICD-topology.zip	transitions.csv	Data	8	937 B
SYNTHETIC_ICD-topology.zip	topology.graphml	Check	—	2.7 KiB
SYNTHETIC_ICD-topology.zip	metadata.json	Metadata	—	403 B
SYNTHETIC_ICD-topology.zip	SHA256SUMS.txt	Integrity	—	321 B
SYNTHETIC_ICD-stable-site.zip	sites.csv	Data	8	1.3 KiB
SYNTHETIC_ICD-stable-site.zip	site_cells.csv	Data	16	1.6 KiB
SYNTHETIC_ICD-stable-site.zip	site_rf_neighbours.csv	Context	8	975 B
SYNTHETIC_ICD-stable-site.zip	motion.csv	Data	16	966 B
SYNTHETIC_ICD-stable-site.zip	shadow_episodes.csv	Data	2	643 B
SYNTHETIC_ICD-geometry.zip	geometry_cells.csv	Data	5	854 B
SYNTHETIC_ICD-geometry.zip	geometry_edges.csv	Data	8	1.2 KiB
SYNTHETIC_ICD-geometry.zip	mobility_trips.csv	Data	4	571 B
SYNTHETIC_ICD-geometry.zip	geometry.graphml	Check	—	3.7 KiB
SYNTHETIC_ICD-geometry.zip	metadata.json	Metadata	—	849 B

B. Validation issues

Table 10. Issues grouped by file

File	Code	Severity	Rows	Description and examples
SYNTHETIC_ICD-topology.zip	SCOPE_AGGREGATE	Info	1	Cumulative aggregate: totals per route and last time seen. It contains neither individual handovers nor the period the counters cover.
SYNTHETIC_ICD-stable-site.zip	EXTRA_COLUMNS	Info	1	Columns not interpreted, kept as they are: evidence_type, radio, arfcn, pci, mcc_metadata, mnc_metadata, tac_metadata, observations, distinct_days, first_seen_ms, last_seen_ms.
SYNTHETIC_ICD-stable-site.zip	SCOPE_OVERLAP	Info	1	Each place is recorded under up to 4 overlapping site keys: per-site counters are not added up as unique observations.
SYNTHETIC_ICD-geometry.zip	SCOPE_AGGREGATE	Info	1	Aggregate of routes and trips without coordinates; trip_count may include trips already pruned (older than 90 days).
SYNTHETIC_ICD-geometry.zip	TRIPS_DISCARDED	Info	1	1 CLOSED trip does not meet the validity rule (movement and 3 cells): ICdetector discards them but still exports them as CLOSED.

C. Evidence references by episode

Each row is identified by file and physical line (the header is line 1) so it can be found in the original. Coordinates with visual rounding (annex F).

Table 11. E1: rows with a signal and the row before each one; the rest of the interval (10 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:1278	2026-09-25 06:22:55	001-01-3001-1000980-LTE	-99	OK	0.009, 0.021
SYNTHETIC_patrones_historial.csv:1277	2026-09-25 06:23:20	001-01-3001-1000069-LTE	-95	[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)	0.009, 0.021

Table 12. E2: rows with a signal and the row before each one; the rest of the interval (10 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:1235	2026-09-25 12:01:47	001-01-3002-1000544-LTE	-91	OK	0.012, 0.021
SYNTHETIC_patrones_historial.csv:1234	2026-09-25 12:02:12	001-01-3002-1000545-LTE	-94	[sub-umbral] Downgrade de banda forzado (2620MHz->925MHz, B7->B8)	0.008, 0.019

Table 13. E3: rows with a signal and the row before each one; the rest of the interval (10 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:1228	2026-09-25 12:32:42	`001-01-3002-1000542-LTE`	-93	OK	0.009, 0.022
SYNTHETIC_patrones_historial.csv:1227	2026-09-25 12:33:07	`001-01-3002-1000543-LTE`	-89	[sub-umbral] Downgrade de banda forzado (2620MHz->791MHz, B7->B20)	0.010, 0.021

Table 14. E4: rows with a signal and the row before each one; the rest of the interval (7 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:856	2026-09-27 16:37:14	`001-01-3001-1000980-LTE`	-101	OK	0.009, 0.019
SYNTHETIC_patrones_historial.csv:855	2026-09-27 16:48:53	`001-01-3001-1000070-LTE`	-81	[sub-umbral] Potencia anómala vs historial (+19dB)	0.012, 0.021

Table 15. E5: rows with a signal and the row before each one; the rest of the interval (8 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_ICD-stable-site.zip!shadow_episodes.csv:2	2026-09-27 21:14:00	`001-01-1099-9999001-LTE`	—	Stable-Site: NEW_SERVING_WHILE_STATIC_AT_MATURE_SITE	—

Table 16. E6: rows with a signal and the row before each one; the rest of the interval (80 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:501	2026-09-29 13:53:20	`001-01-3003-1000953-LTE`	-106	OK	0.032, 0.040
SYNTHETIC_patrones_historial.csv:500	2026-09-29 13:53:44	`001-01-3003-1000953-LTE`	-103	[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)	0.032, 0.041
SYNTHETIC_patrones_historial.csv:498	2026-09-29 13:54:12	`001-01-3002-1000060-LTE`	-94	OK	0.030, 0.042
SYNTHETIC_patrones_historial.csv:496	2026-09-29 13:54:36	`001-01-3003-1000953-LTE`	-112	[sub-umbral] Identidad RF inestable: misma Cell ID alternando PCI en la misma portadora (posible clon)	0.029, 0.041

Table 17. E7: rows with a signal and the row before each one; the rest of the interval (87 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:488	2026-09-29 14:00:17	`001-01-3002-1000060-LTE`	-100	OK	0.009, 0.019
SYNTHETIC_patrones_historial.csv:487	2026-09-29 14:00:42	`001-01-3002-1000061-LTE`	-92	[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)	0.010, 0.019

Table 18. E8: rows with a signal and the row before each one; the rest of the interval (97 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:480	2026-09-29 14:05:32	`001-01-3002-1000542-LTE`	-94	OK	0.012, 0.020
SYNTHETIC_patrones_historial.csv:479	2026-09-29 14:05:57	`001-01-3002-1000545-LTE`	-90	[sub-umbral] Downgrade de banda forzado (2620MHz->925MHz, B7->B8)	0.009, 0.020
SYNTHETIC_patrones_historial.csv:460	2026-09-29 14:18:09	`001-01-3002-1000060-LTE`	-99	OK	0.009, 0.019
SYNTHETIC_patrones_historial.csv:459	2026-09-29 14:18:24	`001-01-3002-1000545-LTE`	-83	[sub-umbral] Downgrade de banda forzado (1805MHz->925MHz, B3->B8)	0.011, 0.019
SYNTHETIC_patrones_historial.csv:450	2026-09-29 14:26:39	`001-01-3003-1000989-LTE`	-106	OK	0.032, 0.039
SYNTHETIC_patrones_historial.csv:449	2026-09-29 14:26:58	`001-01-3002-1000545-LTE`	-84	[sub-umbral] Downgrade de banda forzado (1805MHz->925MHz, B3->B8)	0.009, 0.019

Table 19. E9: rows with a signal and the row before each one; the rest of the interval (84 rows within ±30 min) is in the original

Reference	Local time	Identity	dBm	ICdetector reason (original)	Position
SYNTHETIC_patrones_historial.csv:462	2026-09-29 14:17:25	`001-01-3002-1000544-LTE`	-99	OK	0.010, 0.018
SYNTHETIC_patrones_historial.csv:461	2026-09-29 14:17:50	`001-01-3002-1000543-LTE`	-90	[sub-umbral] Downgrade de banda forzado (2620MHz->791MHz, B7->B20)	0.008, 0.021
SYNTHETIC_patrones_historial.csv:438	2026-09-29 14:33:46	`001-01-3002-1000060-LTE`	-95	OK	0.031, 0.041
SYNTHETIC_patrones_historial.csv:437	2026-09-29 14:34:03	`001-01-3002-1000543-LTE`	-83	[sub-umbral] Downgrade de banda forzado (1805MHz->791MHz, B3->B20)	—

D. Statistical description of the history

Table 20. Serving identities with the most rows (12 of 12)

Identity	Rows	% rows	First (local)	Last (local)
001-01-3001-1000070-LTE	876	66.3%	2026-09-24 23:46:33	2026-10-02 03:11:05
001-01-3002-1000060-LTE	84	6.4%	2026-09-25 08:46:04	2026-10-01 19:22:37
001-01-3003-1000989-LTE	68	5.1%	2026-09-25 09:28:15	2026-10-01 18:50:47
001-01-3002-1000542-LTE	67	5.1%	2026-09-25 10:52:47	2026-10-01 20:47:22
001-01-3002-1000544-LTE	65	4.9%	2026-09-25 09:48:36	2026-10-01 20:36:40
001-01-3001-1000069-LTE	56	4.2%	2026-09-25 06:23:20	2026-10-01 20:26:05
001-01-3001-1000071-LTE	48	3.6%	2026-09-25 09:23:14	2026-10-01 14:34:39
001-01-3001-1000980-LTE	45	3.4%	2026-09-25 06:22:55	2026-10-01 19:01:22
001-01-3002-1000545-LTE	4	0.3%	2026-09-25 12:02:12	2026-09-29 14:26:58
001-01-3003-1000953-LTE	4	0.3%	2026-09-29 13:53:20	2026-09-29 13:54:36
001-01-3002-1000543-LTE	3	0.2%	2026-09-25 12:33:07	2026-09-29 14:34:03
001-01-3002-1000061-LTE	1	0.1%	2026-09-29 14:00:42	2026-09-29 14:00:42

Table 21. Recorded power (dBm) by technology; technologies are not combined

Technology	Quantity	n	Min.	P10	Median	P90	Max.
LTE	getDbm(), RSRP on current Android	1,321	-118	-105	-99	-92	-81

ICdetector results	OK 1,309 · sub-threshold 12 · reason without prefix 0 · SecurityScore < 100 in 12 rows
AnomalyConfidence	maximum 9.3: an uncalibrated internal score, not a probability
External verification	NOT_FOUND 941; PENDING 65; VERIFIED 315
TA consistency	0 rows with TAMeters different from the documented conversion
Duplicates in the file	0 identical; 0 with the same time and identity; none removed
Interval with no records	threshold 60 min

E. Rules and thresholds

- Grouping: same cell and same pattern type within <= 30 min; groups of episodes within <= 15 min (without merging them); Radio within <= 5 min.
- Observation moment: signals within <= 60 s are counted once. Persistence: >= 3 moments over >= 10 min. Recurrence: same cell and pattern on >= 2 days.
- LTE band: Bands column or EARFCN (3GPP TS 36.101); compared with the previous row within <= 10 min.
- Power: median of the same cell outside ±30 min of the episode, with >= 5 rows; significant difference >= 10 dB.
- Evidence families: collector (history, forensic case, Stable-Site) and phone state (Radio). Radio is the only separate family; a coincidence with Radio is reported as context and does not raise the priority.
- Interval with no records: 60 min or more between rows with an exact time. Valid ranges and sentinels: docs/analysis-rules.md.
- Versions: cellcase-rules/3, episodes/7, readers/2.4.1, cross/4.

F. Privacy and rounding

- Coordinates: rounded to 3 decimals (around 100 m).
- Cell identifiers: complete.
- Powers and percentages are shown with the rounding stated in each table; the calculations use the original values.
- The original files kept in the app are not modified.